

10.18 (a) From Eq. (3.183), the effective electric dipole moment from the circular hole is

$$\vec{p} = -\frac{4\epsilon_0}{3}(\vec{E}_0 - \vec{E}_1)a^3, \text{ where } \vec{E}_0 = 0 \text{ and } \vec{E}_1 = \vec{E}_0 e^{i\omega t} S_0,$$

$\vec{p} = \frac{4\epsilon_0}{3}\vec{E}_0 a^3 e^{-i\omega t}$. By Eq. (9.19), the electric field from the dipole moment in the radiation zone is

$$\vec{E}_1 = \frac{Z_0 c k^2}{4\pi} \left[(\vec{n} \times \vec{p}) \times \vec{n} \right] \frac{e^{ikr}}{r} = \frac{e^{ikr-i\omega t}}{3\pi r} Z_0 c \epsilon_0 k^2 a^3 \left[(\vec{n} \times \vec{E}_0) \times \vec{n} \right].$$

Since $Z_0 = \sqrt{\mu_0/\epsilon_0}$ and $c = 1/\sqrt{\mu_0\epsilon_0}$, we have

$$\vec{E}_1 = \frac{e^{ikr-i\omega t}}{3\pi r} k^2 a^3 \left[(\vec{n} \times \vec{E}_0) \times \vec{n} \right].$$

For the magnetic field component, by Eq. (5.131), the magnetic dipole moment is

$$\vec{m} = -\frac{8a^3}{3\mu_0} \vec{B}_0 e^{-i\omega t}. \text{ Notice the sign reversal due to the location of the magnetic field.}$$

Then, by Eq. (9.36), the electric field from the magnetic dipole moment is

$$\vec{E}_2 = -\frac{Z_0}{4\pi} k^2 (\vec{n} \times \vec{m}) \frac{e^{ikr}}{r} = \frac{e^{ikr-i\omega t}}{3\pi r} \frac{2Z_0}{\mu_0} k^2 a^3 (\vec{n} \times \vec{B}_0) = \frac{e^{ikr-i\omega t}}{3\pi r} 2c k^2 a^3 (\vec{n} \times \vec{B}_0).$$

Thus, the electric field is

$$\vec{E} = \vec{E}_1 + \vec{E}_2 = \frac{e^{ikr-i\omega t}}{3\pi r} k^2 a^3 \left[2c \frac{\vec{k}}{k} \times \vec{B}_0 + \frac{\vec{k}}{k} \times (\vec{E}_0 \times \frac{\vec{k}}{k}) \right],$$

with $\vec{n} = \vec{k}/k$.

(b) The time-averaged power radiated per unit solid angle is

$$\frac{dp}{d\Omega} = \frac{1}{2Z_0} r^2 |\vec{E}|^2 = \frac{1}{18\pi^2 Z_0} k^4 a^6 \left| 2c \frac{\vec{k}}{k} \times \vec{B}_0 + \frac{\vec{k}}{k} \times (\vec{E}_0 \times \frac{\vec{k}}{k}) \right|^2.$$

For $\vec{k} = k(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$, $\vec{B}_0 = B_0(0, 1, 0)$, $\vec{E}_0 = E_0(0, 0, 1)$, we have

$$\frac{\vec{k}}{k} \times \vec{B}_0 = B_0(-\cos\theta, 0, \sin\theta\cos\phi),$$

$$\frac{\vec{k}}{k} \times (\vec{E}_0 \times \frac{\vec{k}}{k}) = E_0(\sin\theta\cos\theta\cos\phi, \sin\theta\cos\theta\sin\phi, \sin^2\theta)$$

$$\text{and } \frac{dp}{d\Omega} = \frac{1}{18\pi^2 Z_0} k^4 a^6 \left[4c^2 B_0^2 (\cos^2\theta + \sin^2\theta\cos^2\phi) + E_0^2 \sin^2\theta - 4c E_0 B_0 \sin\theta\cos\theta\cos\phi \right]$$

To calculate the total power transmitted through the hole, we only need to consider the $z > 0$ half-space and the angular integrals are

$$\int_0^{\pi/2} \sin\theta d\theta \int_0^{2\pi} d\phi \sin^2\theta = 2\pi \int_0^1 (1 - \cos^2\theta) d(\cos\theta) = 2\pi \cdot \frac{2}{3} = \frac{4\pi}{3}.$$

$$\int_0^{\pi/2} \sin\theta d\theta \int_0^{2\pi} d\phi \cos^2\theta = \frac{2\pi}{3}, \quad \int_0^{\pi/2} \sin\theta d\theta \int_0^{2\pi} d\phi \sin^2\theta \cos^2\phi = \frac{2\pi}{3}$$

$$\int_0^{\pi/2} \sin\theta d\theta \int_0^{2\pi} d\phi \sin\theta \cos 2\theta \cos\phi = 0.$$

Therefore,
$$P = \int_0^{\pi/2} \sin\theta d\theta \int_0^{2\pi} d\phi \frac{dP}{dn} = \frac{1}{18\pi Z_0} k^4 a^6 \cdot \frac{4\pi}{3} (4C^2 B_0^2 + E_0^2)$$

$$= \frac{2}{27\pi Z_0} k^4 a^6 (4C^2 B_0^2 + E_0^2)$$